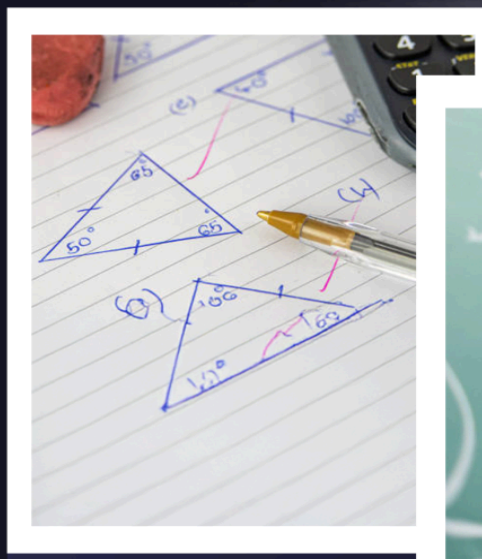
testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Simplification



Simplification

Surds and indices

Rules of surds:

- Surds cannot be added. $\sqrt{a} + \sqrt{b} \neq \sqrt{(a+b)}$
- Surds cannot be subtracted. $\sqrt{a} - \sqrt{b} \neq \sqrt{(a-b)}$
- Surds can be multiplied. $\sqrt{a} \times \sqrt{b} = \sqrt{(a \times b)}$
- Surds can be divided. $\sqrt{a} / \sqrt{b} = \sqrt{(a/b)}$
- Surds can be written in exponential form. $\sqrt{a} = a^{1/2}$

Laws of Surds:

$$\sqrt[n]{ab} = \sqrt[n]{a}\sqrt[n]{b}$$
$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$
$$\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$$
$$\sqrt[n]{a^n} = a \text{ if } n \text{ is odd}$$
$$\sqrt[n]{a^n} = |a| \text{ if } n \text{ is even}$$
$$a^{m/n} = (\sqrt[n]{a})^m = \sqrt[n]{a^m}$$

Laws of Indices:

- $a^m \times a^n = a^{m+n}$
- $a^m \div a^n = a^{m-n}$
- $(a^m)^n = a^{m \times n}$
- $a^0 = 1$
- $a^1 = a$
- $a^{-m} = \frac{1}{a^m}$

Bodmas Rule

VBODMAS stands for -

- V - Vinculum or bar
- B - Brackets
- O - Order of powers or roots
- D - Division
- M - Multiplication
- A - Addition
- S – Subtraction

Percentage

Calculation of a % of b:

⇒ (a/100)* b = (a* b)/100

⇒ a% of b = b% of a

Percentage to Fraction:

1%	=	1/100	25%	=	1/4	80%	=	4/5
2%	=	1/50	33.33%	=	1/3	83.33%	=	5/6
4%	=	1/25	37.50%	=	3/8	87.50%	=	7/8
5%	=	1/20	40%	=	2/5	100%	=	1
8.33%	=	1/12	50%	=	1/2	120%	=	6/5
10%	=	1/10	60%	=	3/5	125%	=	5/4
12.50%	=	1/8	62.50%	=	5/8	133.33%	=	4/3
16.67%	=	1/6	66.67%	=	2/3	150%	=	3/2
20%	=	1/5	75%	=	3/4	175%	=	7/4

Fraction to Percentage:

$\frac{1}{2}=50\%$	$\frac{1}{7}=14\frac{2}{7}\%$	$\frac{1}{12}=8\frac{1}{3}\%$	$\frac{1}{25}=4\%$
$\frac{1}{3}=33\frac{1}{3}\%$	$\frac{1}{8}=12\frac{1}{2}\%$	$\frac{1}{13}=7\frac{9}{13}\%$	$\frac{1}{30}=3\frac{1}{3}\%$
$\frac{1}{4}=25\%$	$\frac{1}{9}=11\frac{1}{9}\%$	$\frac{1}{14}=7\frac{1}{7}\%$	$\frac{1}{40}=2\frac{1}{2}\%$
$\frac{1}{5}=20\%$	$\frac{1}{10}=10\%$	$\frac{1}{15}=6\frac{2}{3}\%$	$\frac{1}{50}=2\%$
$\frac{1}{6}=16\frac{2}{3}\%$	$\frac{1}{11}=9\frac{1}{11}\%$	$\frac{1}{20}=5\%$	$\frac{1}{100}=1\%$

Approx value

Rules of Approximation:

- If a number has digits to the right of the decimal less than 5, then just drop the digits to the right of the decimal. The number so obtained will be the approximated value.
- If a number has digits to the right of the decimal more than 5, then just drop the digits to the right of the decimal and raise the remaining number by '1'.The number so obtained will be the approximated value.